

ARTICLE

Long-term marine protection enhances kelp forest ecosystem stability

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Abstract

Trophic downgrading destabilizes ecosystems and can drive large-scale shifts in ecosystem state. While restoring predatory interactions in marine reserves can reverse anthropogenic-driven shifts, empirical evidence of increased ecosystem stability and persistence in the presence of predators is scant. We compared temporal variation in rocky reef ecosystem state in New Zealand's oldest marine reserve to nearby fished reefs to examine whether protection of predators led to more persistent and stable reef ecosystem states in the marine reserve. Contrasting ecosystem states were found between reserve and fished sites, and this persisted over the 22-year study period. Fished sites were predominantly urchin barrens but occasionally fluctuated to short-lived turfs and mixed algal forests, while reserve sites displayed unidirectional successional trajectories toward stable kelp forests (*Ecklonia radiata*) taking up to three decades following protection. This provides empirical evidence that long-term protection of predators facilitates kelp forest recovery, resists shifts to denuded alternate states, and enhances kelp forest stability.

KEYWORDS

algal turf, alternate states, kelp forest, marine protected area (MPA), marine reserve, persistence, predatory interactions, resilience, shallow temperate reefs, urchin barrens

INTRODUCTION

Overfishing and other anthropogenic impacts on marine ecosystems have led to trophic downgrading and large-scale shifts in ecosystems globally (Estes et al., 2011; Jackson et al., 2001; Lotze et al., 2006; McCauley et al., 2015). Due to the ubiquitous and primal impacts of fishing on global oceans, marine protected areas (MPAs) are often seen as an effective management tool to reverse and mitigate the impacts of fishing and increase resilience to other stressors such as climate change (Duarte et al., 2020; Roberts et al., 2017; Sala & Giakoumi, 2018; Tittensor et al., 2019). In addition to allowing recovery of

species directly impacted by harvesting, MPAs can restore lost trophic interactions (e.g., trophic cascades) and indirectly lead to wider ecosystem recovery and increased ecosystem health (Babcock et al., 2010). Stability is a key attribute of ecosystem health (Costanza, 1992; Rapport et al., 1998), and from a conservation and management perspective, it is central to ensuring that efforts taken in marine protection lead to favorable persistent long-term outcomes. While it has been proposed that the reestablishment of predatory interactions in marine reserves could increase ecosystem stability and resilience, empirical evidence is rare (Bates et al., 2014; Fraschetti et al., 2013; Mellin et al., 2016). Understanding the

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capacity of marine protection to enhance ecosystem stability is particularly important for managing ecosystems where anthropogenic disturbance erodes ecosystem resilience and can lead to large-scale phase shifts or switches to alternative stable states (Folke et al., 2004; Scheffer et al., 2001). Long-term studies are consequently crucial to understanding the time frames and successional pathways involved in restoring ecosystem resilience and stability.

Shallow temperate reef ecosystems are typically dominated by macroalgal forests (mainly kelp, order Laminariales, or Fucales) that provide habitat for myriad species and vital ecosystem functions and services (Steneck & Johnson, 2013). However, overfishing of species at high trophic levels has been shown to disrupt predatory interactions and generate trophic cascades, resulting in the proliferation of sea urchins and consequent destructive grazing of macroalgal forests to form an urchin dominated state (termed “urchin barren”) devoid of large macroalgae (Filbee-Dexter & Scheibling, 2014; Steneck et al., 2002). This shift from kelp forest to urchin barren has been documented in many temperate regions worldwide (Filbee-Dexter & Scheibling, 2014; Ling et al., 2015; Sala et al., 1998) and has been classified as a catastrophic phase shift (Ling et al., 2009). The recovery of predators within MPAs, has been shown to restore predator–prey interactions and reverse these catastrophic phase shifts (Babcock et al., 1999; Edgar et al., 2017; Guidetti, 2006; Sangil et al., 2012; Shears & Babcock, 2003), but this can take decades from the time fishing has ceased (Babcock et al., 2010; Shears & Babcock, 2003). Various stabilizing feedback mechanisms maintain both kelp forest and urchin barren states (Filbee-Dexter & Scheibling, 2014; Ling et al., 2015), leading to a hysteresis effect between the two states (Petraitis & Latham, 1999). While urchin barrens may display a high degree of stability and persist over decades (Norderhaug & Christie, 2009; Simenstad et al., 1978), urchin populations can exhibit a high degree of temporal variability in relation to extrinsic stressors, which can also lead to fluctuation in the extent of barrens in the absence of predators (Behrens & Lafferty, 2004; Lafferty, 2004; Williams et al., 2021).

Over the past decade, reports of shifts from kelp forests to an algal turf state have also emerged from many temperate reefs (Filbee-Dexter & Wernberg, 2018). This shift is associated with various stressors, including climatic variation, eutrophication and sedimentation that impact kelp and facilitate algal turfs (Connell, 2003; Connell et al., 2008; Feehan et al., 2019; Filbee-Dexter et al., 2016; Moy & Christie, 2012; Rilov et al., 2018; Vergés, Tomas, et al., 2014; Wernberg et al., 2016). The term “turf” generally refers to low-lying, mat-forming or

grass-like algae that occupy the reef (Connell et al., 2014). It may comprise of a monospecific or multispecific assemblage, including perennial and/or ephemeral species that may vary substantially in identity and morphology (Connell et al., 2014). Various types of turfs have shared functions that help maintain algal turf state: some have been shown to trap sediment, inhibit kelp recruitment and increase dislodgment of kelp plants that are attached to turf (Burek et al., 2018; Feehan et al., 2019; Filbee-Dexter et al., 2016; Gorgula & Connell, 2004; Kennelly, 1987; O'Brien & Scheibling, 2018). The presence of such feedbacks generally infers that turfs represent a persistent state (Filbee-Dexter & Wernberg, 2018). This appears to be the case in systems where certain environmental conditions and/or increasing anthropogenic disturbance facilitates turfs and consequently inhibits kelp recovery. Unlike the recovery of kelp forests from urchin barrens, the recovery from an algal turf state will likely require the reversal of the environmental conditions (Filbee-Dexter & Wernberg, 2018). To our knowledge, there has been only one example of a reverse shift from algal turfs to kelp forests (sugar kelp *Saccharina latissimi*) reported in the literature (Christie et al., 2019). That study suggested that algal turfs may constitute a (prolonged) transient state rather than an alternate stable state as was previously thought (Moy & Christie, 2012).

In New Zealand's oldest marine reserve—Cape Rodney–Okakari Point (or Leigh) Marine Reserve (est. 1977)—shallow reef areas that were previously urchin barrens have been replaced by kelp forests and algal turfs after ~16–22 years of no-take protection (Babcock et al., 1999, 2010; Leleu et al., 2012; Parsons et al., 2004; Shears & Babcock, 2002, 2003). These changes have been attributed to density and behaviorally mediated declines in grazing activity of sea urchins (“kina”, *Evechinus chloroticus*) in response to the increase in size and abundance of their main predators: snapper *Pagrus auratus* and spiny lobster *Jasus edwardsii* (Babcock et al., 1999; Kelly et al., 2000; Shears & Babcock, 2002; Smith et al., 2014; Spyksma et al., 2017; Willis et al., 2003). The long time required for these indirect effects to form was attributed to size-specific predation and the time required for predators to reach an impactful size and reduce grazing urchin densities below the threshold required to maintain barrens (Babcock et al., 2010; Salomon et al., 2010; Shears & Babcock, 2003). In contrast, these predators are rare on adjacent fished reefs, and sea urchin barrens remain common (Babcock et al., 1999; Kelly et al., 2000; Shears & Babcock, 2002, 2003; Smith et al., 2014; Spyksma et al., 2017; Willis et al., 2003).

Recent surveys of the dominant sea urchin predators *P. auratus* and *J. edwardsii* have demonstrated that the contrast in predator size and density between the Leigh

Marine Reserve and adjacent fished reefs has persisted (Allard et al., 2022; Hanns et al., 2022), although there have been declines in *J. edwardsii* within the reserve over the past two decades (LaScala-Gruenewald et al., 2021). In contrast, the differing ecosystem states of shallow reefs between protected and fished sites was last documented in the early 2000s (Babcock et al., 1999; Shears & Babcock, 2002, 2003), and the long-term persistence and stability of these patterns has not been examined. This region has not experienced a high degree of long-term ocean warming (Shears & Bowen, 2017), but a number of factors may have influenced the ecosystem dynamics and stability of reef ecosystems inside and outside the reserve over the past two decades. For example, declines in rock lobster in the reserve (LaScala-Gruenewald et al., 2021) and blooms of benthic dinoflagellates *Ostreopsis siamensis* that impact sea urchins have become common in the area over this period (Shears & Ross, 2009, 2010). Both have the potential to influence sea urchin populations and the extent of urchin barrens at reserve and fished sites. Given the unprecedented changes reported on temperate reefs globally over the past 20 years (Filbee-Dexter & Scheibling, 2014; Filbee-Dexter & Wernberg, 2018; Krumhansl et al., 2016; Smale, 2020; Vergés, Steinberg, et al., 2014; Wernberg et al., 2019), it is pertinent to understand whether the contrasting patterns in ecosystem state previously described inside and outside the Leigh Marine Reserve have persisted, and the role of predators in maintaining and promoting kelp forest stability in the face of other stressors.

In this study, we examine long-term rocky reef monitoring data from inside and outside the Leigh Marine Reserve to determine how the contrasting ecosystem states have persisted over three to four decades of no-take protection and, more broadly, examine how the reestablishment of predatory interactions in long-term MPAs affects ecosystem state and stability. Data on the abundance of kelp (*Ecklonia radiata*), fucoids, turfing algae, and sea urchins (*E. chloroticus*) from reefs inside and adjacent to the Leigh Marine Reserve were used to classify the reef according to four ecosystem states (kelp forests, mixed algal forests, urchin barrens, and algal turfs) and to examine temporal variability in these ecosystem states. Ecosystem stability at reserve sites was compared to nearby fished sites based on the number of shifts between ecosystem states and also examining variability in multidimensional space based on key variables defining each ecosystem state. We hypothesized that the reestablishment of predatory interactions and their maintenance by long-term marine protection would increase stability in the reserve compared to nearby fished sites, where other stressors may lead to greater variability in the absence of predators.

MATERIALS AND METHODS

Study area

The Cape Rodney to Okakari Point (Leigh) Marine Reserve (est. 1977) in northeastern New Zealand is a no-take zone that stretches over an area of 547 ha (Appendix S1). The reserve and surrounding coast have extensive subtidal reef systems that are moderately exposed to waves from the northeast and are characterized by continuous reefs that extend from the intertidal zone to a maximum depth of ~25 m. Subtidal reefs in this area display distinct zonation patterns that differ between fished and reserve reefs (Shears et al., 2008; Shears & Babcock, 2003, 2004a, 2004b). Both fished and reserve reefs have a shallow algal fringe (<2 m depth) dominated by a mixed fucoid and kelp (*Ecklonia radiata*) assemblage, and deeper kelp forests (>~7 m depth) largely comprising monospecific stands of *E. radiata*. At intermediate depths (~2–8 m), fished reefs are dominated by urchin barrens, whereas kelp forests dominate reserve reefs. Algal turf habitats may also be common at these intermediate depths in both fished and reserve areas, and these are predominantly comprised of articulated coral-line algae (*Corallina officinalis*) (Shears & Babcock, 2002). Notably, all monitoring data presented in this study are from a 4 to 6 m depth, which is within this intermediate depth range.

Shallow benthic reef monitoring

Historic information on rocky reef communities in the Leigh Marine Reserve is available from baseline surveys in 1978 (Ayling, 1978), and repeated in the 1990s (Babcock et al., 1999). While these provide valuable information on long-term changes in the reserve, they did not include fished reefs outside the reserve that provide a control for the effect of protection. The monitoring program presented in this study was established in 1999 and includes four sites inside (Te Rere: −36.26487, 174.77657; Schiel's Pool: −36.26418, 174.79597; Waterfall Reef: −36.26812, 174.79977; and Onespot: −36.27445, 174.81362) and four sites outside (Kemps: −36.25973, 174.76587; Rodney South: −36.2809, 174.82052; Nordic Reef: −36.29357, 174.80972; and Mathesons Island: −36.30347, 174.80065) the Leigh Marine Reserve (Shears & Babcock, 2003; Shears & Peleg, 2023; Appendix S1). At these sites, the reefs are gradually sloping from the intertidal to the reef edge, occurring at ~8–15 m depth. From 1999 to 2001, the reefs were surveyed in four depth ranges (<2, 4–6, 7–9, and >10 m), but subsequent sampling has focused on the 4–6 m depth range as this is representative of the depth range

where sea urchins are most abundant and can form barren habitats (Shears et al., 2008; Shears & Babcock, 2003). This depth range has been surveyed sporadically (every 1–3 years) since 2001; note that only fished sites were surveyed in 2003 (Appendix S2).

At each site, five 1-m² quadrats were haphazardly sampled within the 4–6 m depth range (Appendix S2). All sampling was non-destructive and done on SCUBA. Within each quadrat, the density of sea urchins and all large brown macroalgae were recorded (e.g., *E. radiata*, *Carpophyllum* spp., *Cystophora* spp. and *Sargassum sinclairii*; also collectively referred to as “forest-forming macroalgae”). For sea urchins, behavior was recorded as “cryptic” (sea urchins found in holes, cracks, or crevices) or “exposed” (sea urchins are out in the open and have a strong impact on macroalgal communities) (Spyksma et al., 2017). The test diameter of all sea urchins was measured to the nearest 5 mm using an incremented scale bar. The length of all large brown macroalgal fronds was measured to the nearest 5 cm. For *E. radiata*, measurements were taken for both stipe and total length (i.e., from the top of the holdfast to the meristem and the distal end of the lamina, respectively). Individual fucoid plants were considered adults if their total length was ≥ 25 cm. For kelp, plants were considered adults if their stipe length was ≥ 15 cm. The percent cover of crustose coralline algae (CCA), bare rock, turfing algae (e.g., articulated coralline, red foliose), filamentous algae, and sediment was assessed visually in each quadrat. In cases where filamentous algae were recorded as overlaying turf and encrusting algae, such that total cover exceeds 100%, the cover data were standardized to add up to 100% by proportionally scaling each variable to the sum of total benthic % cover and multiplying by 100. See Shears and Peleg (2023) for the data.

Classifying ecosystem state

As outlined above, the reef communities at intermediate depths (~2–8 m) in the study area are characterized by four distinct habitat types (Shears & Babcock, 2003; Appendix S3). These include two habitats dominated by forests of large brown macroalgae (phaeophytes)—kelp forest and mixed algal forests—and two habitats lacking large brown macroalgae—urchin barrens and algal turfs. Kelp forests were dominated by the kelp *E. radiata*, whereas “mixed algal forests” were dominated by a mix of *E. radiata* and fucoids, predominantly belonging to the *Carpophyllum* genus. Because these habitat types represent different states of the reef ecosystem, they are referred to here as “ecosystem states” (sensu Filbee-Dexter & Scheibling, 2014). The ecosystem state at each

site for each year was classified based on the quadrat data using the criteria outlined in Appendix S3. These are largely consistent with Shears and Babcock (2003), but we modified “turfing algae” ecosystem state to include the combined cover of turfing algae, filamentous algae, and sediment (hereafter “turf complex”) to better characterize the sediment trapping nature of filamentous algae and algal turfs (analogous to the “epilithic algal matrix” [EAM]; sensu Tebbett & Bellwood, 2019; Wilson & Bellwood, 1997).

Ecosystem state was classified using data on adult kelp and fucoid density, exposed sea urchin density, encrusting algae and bare rock, and turf complex percent cover (details in Appendix S3). Sites were classified as a kelp forest if they were dominated by large brown macroalgae (mean ≥ 4 adult plants m⁻²), and the mean adult *E. radiata* density was higher than that of adult fucoids. Sites dominated by large brown macroalgae (mean ≥ 4 adult plants m⁻²), with a mean fucoid density that exceeded that of adult *E. radiata*, were classified as a mixed algal forest. Sites with low large brown macroalgal density (mean < 4 adult plants m⁻²), and high mean encrusting cover (combined cover of bare rock, CCA, and other encrusting algae $> 50\%$) were classified as an urchin barren. Sites with low large brown macroalgal density (mean < 4 adult plants m⁻²) and high mean turf complex cover ($> 50\%$) were classified as algal turf.

Assessment of stability of key taxa and ecosystem state

To assess the temporal stability of key taxa that define each ecosystem state at reserve and fished sites, the inverse of the coefficient of variation (1/CV) was calculated for each site across the monitoring years. The calculation was done for exposed sea urchin density (m⁻²), turf complex cover (%), adult fucoid density (m⁻²), and *E. radiata* density (m⁻²). The overall temporal stability (1/CV) of each taxa calculated for each site was compared between reserve and fished sites using a *t*-test.

To quantify the overall impact of protection (or “protection effect”), the difference between reserve and fished sites in mean exposed sea urchin density, turf complex % cover, and forest-forming macroalgal density (i.e., adult fucoids and kelp density, jointly) was calculated for each year. The protection effect on fucoids and kelp was calculated together because they were used collectively as a state variable in the reef ecosystem stability (see below).

To assess the reef ecosystem stability inside and outside the reserve, we quantified the number of shifts

between the four ecosystem states in consecutive sampling years for each site. A comparison of the stability of fished and reserve areas was made using the proportion of shifts throughout the entire study and the mean of the total number of shifts per site. Relative stability was calculated for fished and reserve areas by calculating the proportion of sites that did not undergo shifts in ecosystem states between survey years.

Additionally, the multivariate distance in ecosystem state space was calculated to quantify the extent of change in ecosystem state at each site through time. This was done using a multivariate analysis of three state variables: forest-forming macroalgal density, exposed sea urchin density, and the percent cover of turf complex. These state variables broadly characterize the following ecosystem states: macroalgal forests (including mixed algal forests and kelp forests), urchin barrens, and algal turfs. Based on these three variables, the positions of the sites every survey year were calculated in three-dimensional space. Values of the three state variables were normalized relative to the groups' mean and standard deviation (*z*-score). The Euclidean distance between each two consecutive years and the sum of the distances between all consecutive survey years (i.e., "length of chain") was calculated for each site.

Statistical analysis

Temporal variation in the abundance of sea urchins (cryptic, exposed, and total), turf complex, adult large brown macroalgae (*E. radiata*, fucoids and total), and the variation in state space between consecutive years among fished and reserve sites were tested using univariate analysis of variance by permutation (PERMANOVA, 9999 perms, type III SS; Anderson et al., 2008). The analyses were run on Euclidean distance matrices of the samples (i.e., quadrats at any given site and year) using the fixed factors "Year" and "Status" (fished vs. reserve), and the random factor "Site" (nested within each "Year" and "Status" combination). Only fished sites were sampled in 2003, so this year was omitted from the analysis.

RESULTS

Protection effects on key species and groups

The density of exposed sea urchins was consistently higher at fished sites compared to reserve sites, and this effect was stable over time (Figure 1a, Table 1). Furthermore, the stability of exposed sea urchin density was

significantly higher at fished sites compared to reserve sites, as evident from higher $1/CV$ values ($t_6 = 2.86$, $p = 0.029$, two-tailed *t*-test; Figure 1a). Total urchin density was also consistently higher at fished sites, but the density of cryptic urchins was generally higher at reserve sites (Appendix S4). Within the reserve, the majority of sea urchins were cryptic (58%–100%), whereas the majority of urchins at fished sites were exposed (53%–92%). The contrasting behavior and overall size structure of sea urchin populations between reserve and fished sites remained relatively stable over the monitoring period (Appendix S5). At fished sites, the population was dominated by exposed sea urchins >50 mm, whereas smaller urchins were generally less abundant and cryptic. At reserve sites, urchin populations had a more bimodal size distribution dominated by cryptic individuals, with generally only a low number of large urchins (>~80 mm) exhibiting an exposed behavior.

The cover of turf complex was generally higher at fished sites, but the effect of protection varied over time as indicated by the significant interaction between "Status" and "Year" (Figure 1b; Table 1). Turf complex cover was initially similar between reserve and fished sites, but diverged in the early 2000s as adult brown macroalgal density increased at reserve sites (Figure 1b–d). At fished sites, the mean cover of turf complex fluctuated greatly over time (Figure 1b). Overall, the stability of turf complex cover did not vary significantly among fished and reserve sites, as reflected from their $1/CV$ means that were not significantly different ($t_6 = 1.73$, $p = 0.134$; Figure 1b).

The density of forest-forming macroalgae (adult fucoids and *E. radiata*) was significantly higher in the reserve relative to the fished area, and this effect remained constant over time (Figures 1c,d and 3; Table 1). Adult fucoid density increased substantially at reserve sites from 1999 to 2001 following the decline in exposed sea urchins to zero, but by 2006 *E. radiata* had become the dominant forest-forming macroalgae (Figure 1c,d). Despite an increase in exposed sea urchins in the reserve between 2001 and 2011, forest-forming macroalgal density has remained relatively stable (Figure 1a,c,d). A large increase in *E. radiata* occurred at reserve sites and one fished site (Kemps) in 2019 (Figure 1d). Overall, fucoids did not have a significant difference in their temporal stability between fished and reserve sites, and *E. radiata* had higher stability at reserve compared to fished sites, as reflected by their $1/CV$ means ($t_{3,18} = 1.72$, $p = 0.178$, and $t_6 = 2.85$, $p = 0.029$, for fucoids and *E. radiata*, respectively, two-tailed *t*-test; Figure 1c,d).

Overall, protection had a consistent negative effect on the density of exposed sea urchins throughout the study

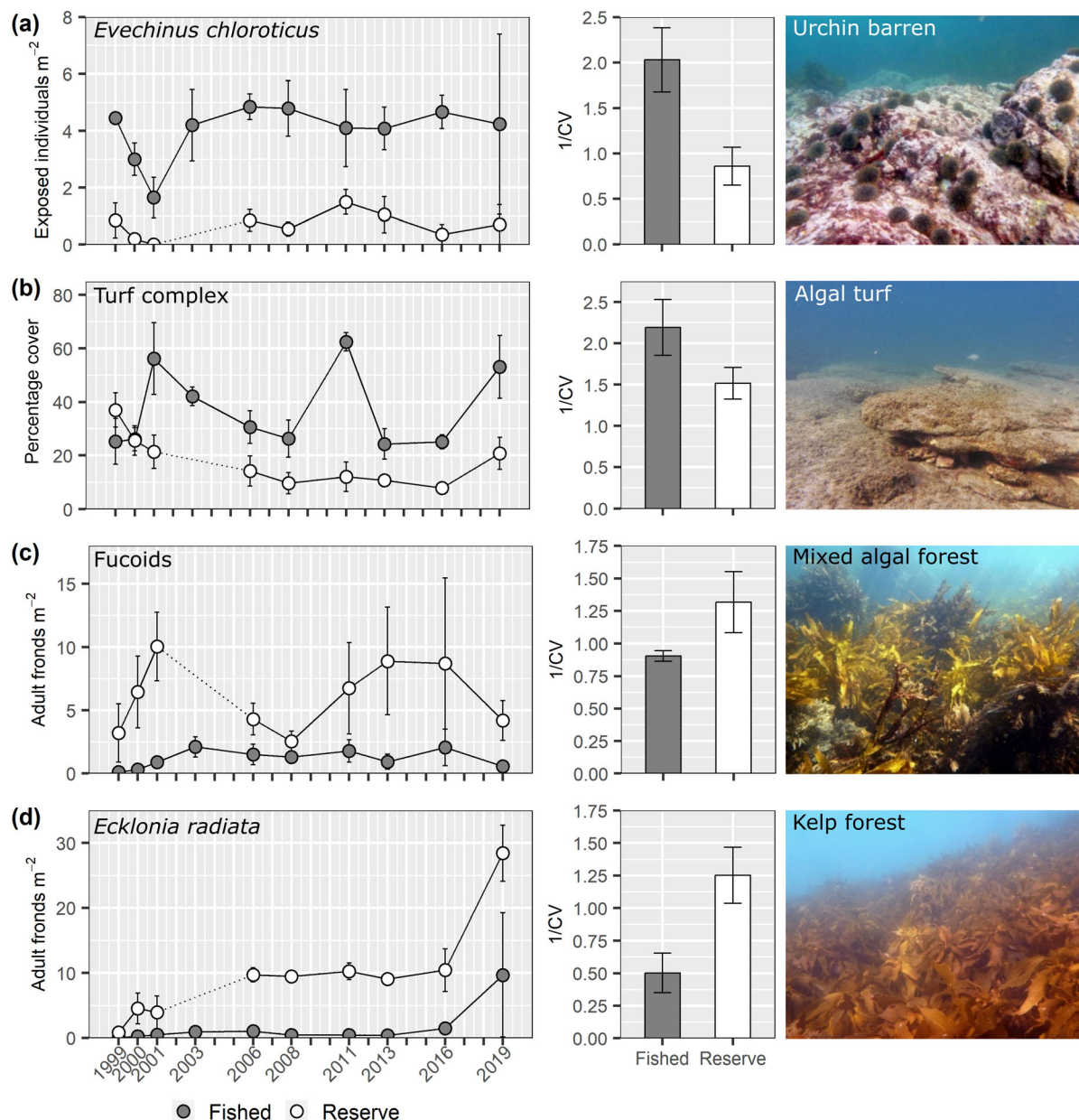


FIGURE 1 Temporal variation between fished and reserve areas in the mean density of (a) “exposed” sea urchins *Evechinus chloroticus*, (b) the percentage cover of turf complex (i.e., the sum cover of turf and filamentous algae and sediment), (c) the density of adult fucoids, and (d) the density of adult *Ecklonia radiata*. The dotted lines show a gap in the time series in 2001–2006 in the reserve sites. The bar graphs to the right estimate the temporal stability of the key taxa shown in (a–d), using the mean inverse of the coefficient of variation (1/CV) of their abundance. The error bars represent the standard error of the sites ($n = 4$). The photos are of urchin barren, algal turf, mixed algal forest, and kelp forest ecosystem states (top to bottom; photos credit: Nick Shears).

(Figure 2). The effect of protection on turf complex was negative throughout the study, except for 1999 when turf complex cover was higher in the reserve, and 2000 when it was similar at reserve and fished sites (Figures 1b and 2). Protection had a consistent positive effect on forest-forming macroalgae throughout the study, but this increased substantially between 1999 and 2001, and then remained relatively stable (Figure 2). Generally, the effect of protection on turf complex and adult brown macroalgae mirrored each other (Figure 2).

As the positive effect on forest-forming macroalgae increases, there was an increasing negative protection effect on turf complex (Figure 2).

Temporal variation in ecosystem state

There was a clear contrast in ecosystem states and the trajectories of sites between reserve and fished sites over the monitoring period (Figure 3). All fished sites were

TABLE 1 Univariate PERMANOVA (9999 unique permutations) results on the fixed terms “Year” and “Status” and the random terms “Site (Year, Status)”.

Variable	Fixed						Random
	Year		Status		Year $\times$ Status		Site (Year, Status)
	F^a	F^b	F^c	F^d	F^e	F^b	F^f
Cryptic sea urchins density (individuals m^{-2})	5.2***		13.2***		0.6		1.9**
Exposed sea urchins density (individuals m^{-2})	0.9		49.1***		0.4		2.9***
Total sea urchins density (individuals m^{-2})	3.1**		8.3**		0.2		2.6***
Turf complex (percent cover)	3.3**		48.0***		3.6**		4.0***
Fucoids density (adult fronds m^{-2})	0.7		19.5***		0.6		2.5***
Ecklonia radiata density (adult fronds m^{-2})	7.9***		36.9***		1.7		8.3***
Total forest-forming macroalgal density (adult fronds m^{-2})	4.8***		61.6***		1.1		4.3***
Variation in state space between consecutive years	8.1***		22.8***		3.0**		

Note: Data from 2003 was omitted from the analysis. Bold indicates significance, and asterisks indicate the level of significance ($p < 0.1$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$).

^a $F_{5,291}$.

^b $F_{7,48}$.

^c $F_{1,291}$.

^d $F_{1,48}$.

^e $F_{8,291}$.

^f $F_{54,291}$.

initially classified as urchin barrens in 1999, and periodically fluctuated to algal turf and occasionally mixed algal forest (Figure 3). The only exception was Kemps which shifted to a kelp forest state in 2019 (Figure 3). The position in state space of reserve sites was proximate to fished sites at the start of the monitoring period, with one classified as mixed algal forest (Te Rere), one as algal turf (Waterfall Reef), and two as urchin barrens (Schiel's Pool, and Onespot) (Figure 3). Waterfall Reef and Schiel's Pool shifted to mixed algal forest state in 2000, and Onespot in 2001 (Figure 3). All reserve sites shifted from mixed algal forest to kelp forest in the early 2000s and have persisted as kelp forests since, excluding Te Rere (Figure 3). Te Rere has predominantly been classified as mixed algal forest and occasionally kelp forest over time (Figure 3). Throughout this study, 20 shifts between states were recorded at the fished sites, compared to only 11 in the reserve (excluding 2003; Figure 3). All of the shifts in the reserve were unidirectional from urchin barrens or algal turfs to mixed algal forest and kelp forests (except at Te Rere), whereas at fished sites they fluctuated back and forth between urchin barrens and algal turfs or mixed algal forests. The proportion of shifts was significantly lower in the reserve ($\chi^2 = 4.00$, $p = 0.023$, one-tailed two-proportions test), where the mean number of shifts per site was significantly lower compared to the fished area ($t_6 = 2.64$, $p = 0.019$, one-tailed t -test; Figures 3 and 4a). No shifts

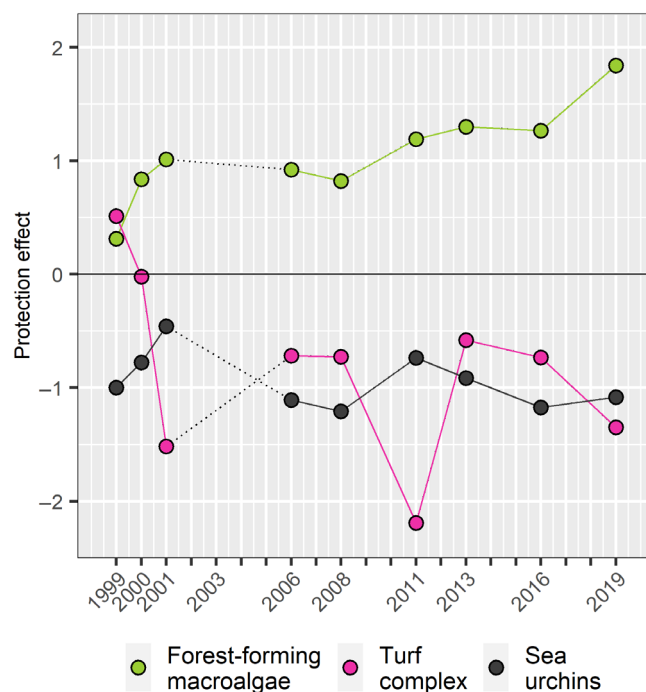


FIGURE 2 The relative effect of no-take protection on forest-forming macroalgae, turf complex (i.e., the sum cover of turf and filamentous algae and sediment), and exposed sea urchins (*Evechinus chloroticus*) density. Calculation was based on normalized values (using z-scores). Values >0 indicate positive effect and <0 , negative. The dotted lines show a gap in the time series in 2001–2006.

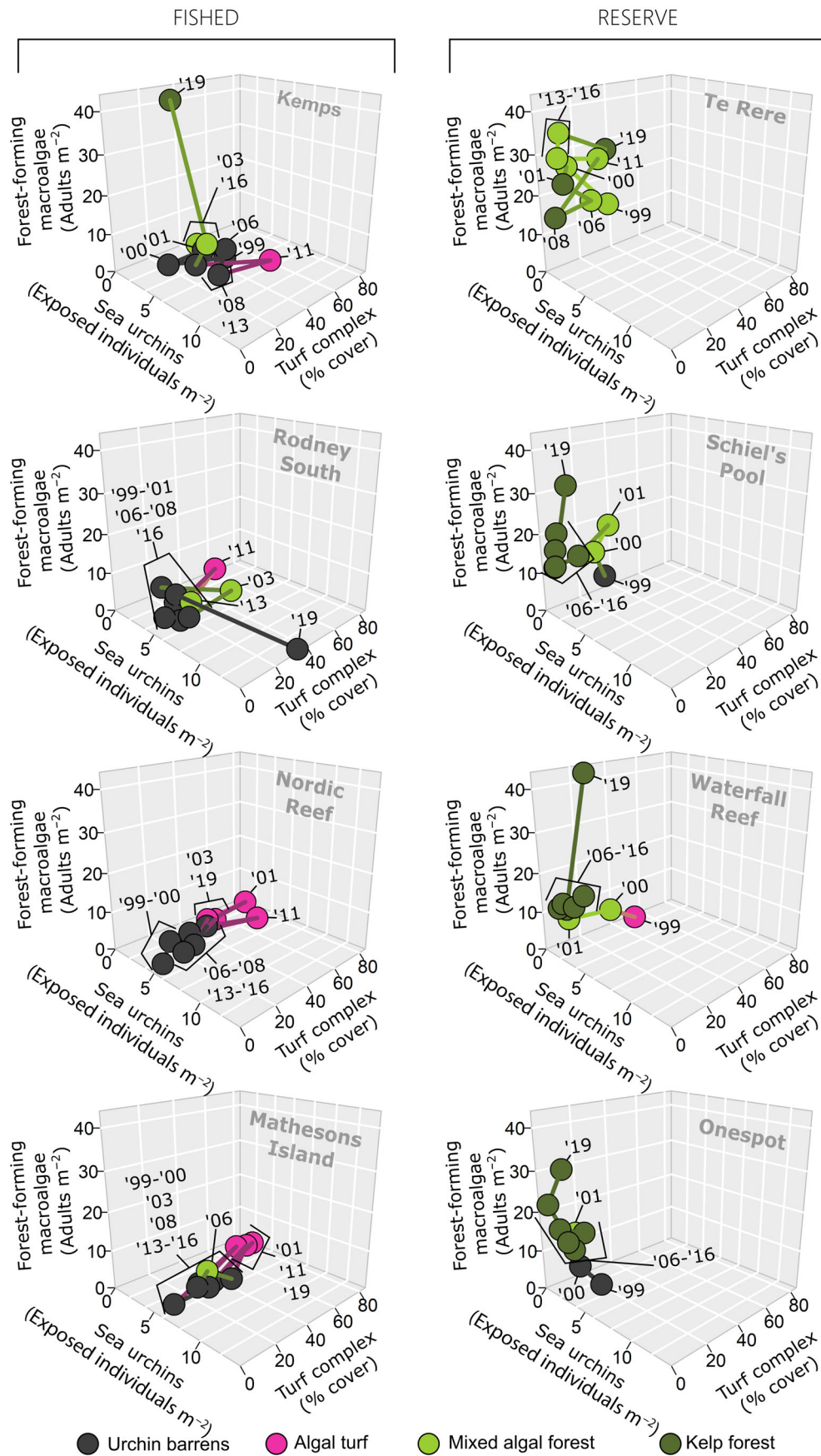


FIGURE 3 Legend on next page.

were detected in the fished area in 2000 where all sites remained in a barren state (Figure 4b).

Throughout this study, the overall sites' mean variation in state space (i.e., "length of chain") was significantly greater in the fished area than in the reserve ($t_6 = 11.22$, $p < 0.001$, one-tailed t -test; Figures 3 and 4c). The distance between consecutive years in three-dimensional state space was generally greater in the fished area, but the effect of protection varied over time as indicated by the significant interaction between "Status" and "Year" (Figure 4d; Table 1).

DISCUSSION

The results of this study show that long-term protection in the Leigh Marine Reserve has resulted in an ecosystem trajectory towards a kelp forest state, and once established, the kelp forest has remained stable despite subsequent fluctuations in sea urchin density. In contrast, at fished sites, where sea urchin predators are rare, urchin barrens were most prevalent, but reefs fluctuated to short-lived algal turfs and, on a few occasions, to mixed algae or kelp forest. These findings provide empirical evidence that long-term marine protection can increase ecosystem stability of kelp forests.

The stability of kelp forests increased in the reserve through a net negative effect of protection on exposed sea urchin abundance and turf complex cover and a net positive effect on forest-forming macroalgal density, thereby resisting shifts to non-canopy states and facilitating forest recovery (Figure 2). This was also reflected in the overall greater stability of exposed sea urchins density at fished sites compared to reserve sites, and contrastingly, greater stability of *E. radiata* density at reserve sites compared to fished sites. Overall, 65% of the total observed shifts among ecosystem states in our study occurred at fished sites where large predators are rare with comparatively small body sizes (Allard et al., 2022; Hanns et al., 2022; LaScala-Gruenewald et al., 2021), indicating reduced stability compared to reserve sites (Figure 3). Moreover, there was a greater variation in ecosystem state space at fished compared to reserve sites. The differences in stability measures between the reserve and fished areas

show how marine protection and the reestablishment of predatory interactions enhances the stability of kelp forest ecosystems. Similar results have been reported from the Mediterranean Sea and Tasmania, where protected shallow reef community assemblages and biodiversity (respectively) showed lower temporal variability in protected areas where predators are more abundant than adjacent fished reefs (Bates et al., 2014; Fraschetti et al., 2013).

The timing of declines in exposed sea urchins within the reserve, and the subsequent contrast in density and size structure of sea urchins between reserve and fished sites, are consistent with the greater size and abundance of predators in the reserve. The decline in sea urchins and increase in kelp forests was first reported at sites within the reserve in the mid-1990s (Babcock et al., 1999, 2010; Cole & Keuskamp, 1998), prior to the establishment of this monitoring program. The impact of predators on sea urchin population was evident in the bimodal distribution in the reserve compared to a unimodal distribution on fished reefs, with greatest predation in the reserve on mid-size urchins moving from a cryptic to exposed lifestyle (Babcock et al., 2010; Cole & Keuskamp, 1998; Shears & Babcock, 2002). Furthermore, in the reserve only the largest sea urchins (≥ 80 mm) exhibit an exposed lifestyle, with these size classes likely being the least vulnerable to predation. This contrasting size structure and behavior of sea urchins between reserve and fished sites has remained evident throughout the 20-year monitoring period.

When the reserve sites examined in the current study were first surveyed in 1999, they were characterized by a combination of urchin barrens, algal turfs, and mixed algal forests; adult kelp was rare, and on average, exposed sea urchin density was low ($\sim 1 \text{ m}^{-2}$). All of the reserve sites subsequently transitioned to kelp forest by 2006, and all but one (Te Rere) have retained this state since. This recovery of forest-forming macroalgae followed the decline in exposed sea urchin numbers below densities of $\sim 1 \text{ m}^{-2}$, which is estimated to be the density required to maintain barrens (Shears & Babcock, 2003). The successional patterns observed were consistent with expectations based on experimental removal of sea urchins, where filamentous and turfing algae initially increase, followed by an increase in fucoids and then kelp (Andrew & Choat, 1982; Ayling,

FIGURE 3 Temporal variations in ecosystem state, expressed as variations in position in state space at fished (left) and reserve (right) sites, using the variables: (x) exposed sea urchin (*Evechinus chloroticus*) density, (y) turf complex (i.e., the sum percentage cover of turfing and filamentous algae and sediment), and (z) forest-forming macroalgal density. Colors represent different ecosystem states: urchin barren (gray), algal turf (maroon), mixed algal forest (light green), and kelp forest (dark green). Each sample represents the means of five quadrats. The connecting lines show the temporal trajectory in state space. The number of ecosystem state shifts is reflected by the number of changes in color along the timeline. Note: 2003 is missing from the reserve sites.

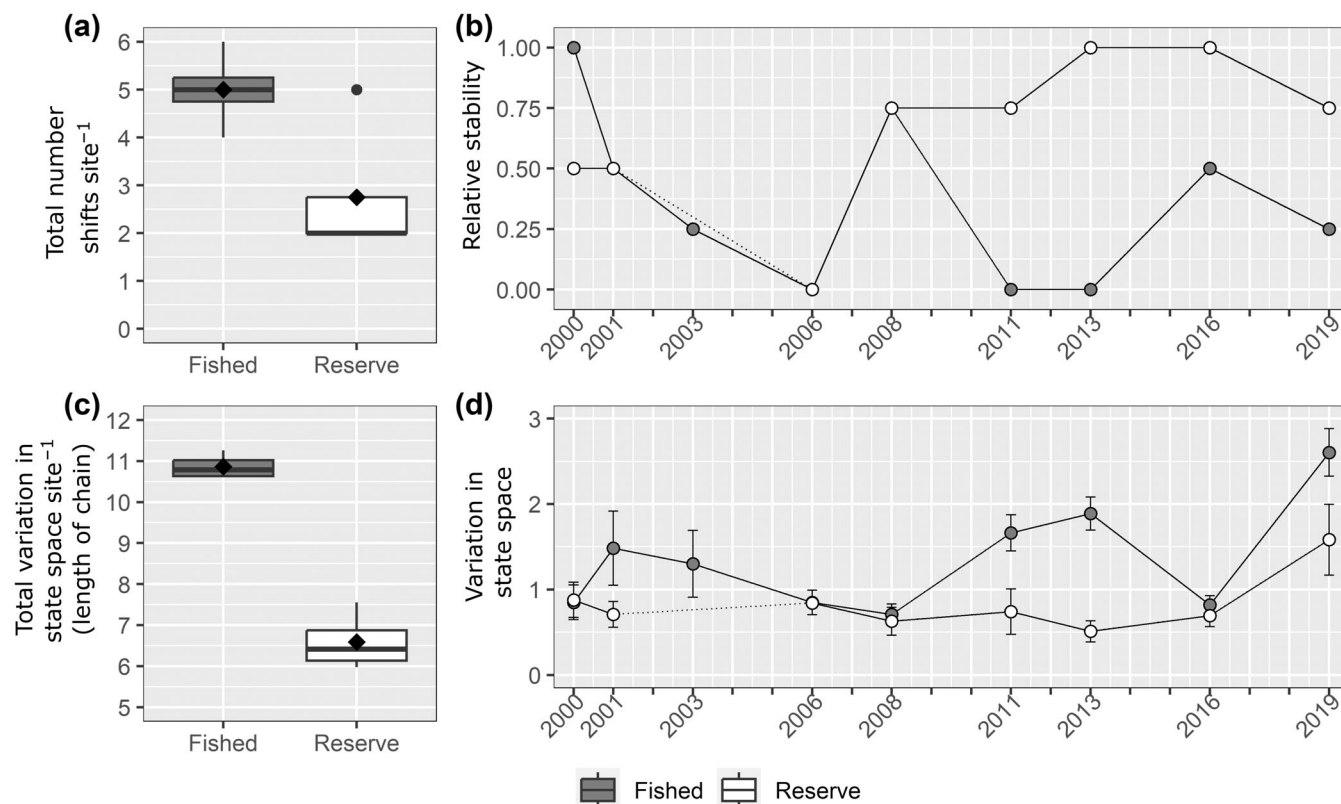


FIGURE 4 Temporal stability of the reef ecosystem between fished and reserve sites: (a) the total number of shifts per site throughout the study; (b) temporal variation in the relative stability, where a value of one indicates that all sites remained stable relative to the previous year and zero indicates that all sites shifted; (c) the total variation in state space of sites throughout the study, that is “length of chain”; (d) the temporal variation in state space quantified by the mean Euclidean distance between each two consecutive monitoring years. The dotted lines in (b) and (d) show a gap in the time series in 2001–2006 in the reserve sites. Data from 2003 are missing from the reserve; therefore, 2003 was excluded from the calculation of (a) and (c). Error bars represent standard errors of sites ($n = 4$).

1976; Shears & Babcock, 2002). While there was variability in the timing of recovery among sites within the marine reserve, the establishment of stable kelp forests took almost three decades from no-take protection initiation (1977–2006), lagging predator recovery by more than two decades. This time lag can be attributed to the time required for predators to reach impactful body size and subsequently reduce exposed urchin densities below the threshold density required to maintain barrens (Babcock et al., 2010; Shears & Babcock, 2003). The reserve site (Te Rere) fluctuated between a mixed algal forest and kelp forest state, in which it was found at the end of the study. The predominance of a mixed algal forest at Te Rere is likely attributed to slightly shallower depth and higher reef heterogeneity compared to other sites, rather than representing an earlier successional stage. The reef below (>5 m depth) the sampling location at Te Rere has persisted as kelp forest since the early 2000’s (N. Shears, unpublished data).

Fished sites tended to follow similar successional patterns in ecosystem state in relation to fluctuations in sea

urchin densities, but only in one case did urchins decline to the point that enabled kelp recovery. This occurred at one fished site (Kemps) where a large decline in sea urchins and an increase in kelp was recorded in 2019. Sampling at this site was originally undertaken near the urchin barren–kelp forest border (Shears & Babcock, 2003) and the barrens at this site have contracted into shallower water (<4 m, N. Shears, unpublished data). While this site is near the reserve boundary (~150 m, Appendix S1), it is heavily fished, and previous research has demonstrated that large predator density drops sharply at the nearby reserve boundary (Hanns & Shears, 2023; Smith et al., 2014; Willis et al., 2003). Therefore, it is unlikely that predator numbers are sufficient at this site to reduce sea urchins’ numbers sufficiently to contribute to this shift from urchin barrens to kelp forests state. The position of the urchin barren–kelp forest boundary has fluctuated at this site over the monitoring period and likely results from other factors influencing sea urchin numbers such as blooms of *Ostreopsis siamensis* (as outlined below).

In general, fished sites have persisted as urchin barrens with intermittent shifts to algal turfs and in some cases mixed algal forests following declines in sea urchin abundance (e.g., 1999–2001). The turf state was relatively common at some fished sites (e.g., Nordic Reef and Mathesons Island), but urchin numbers remained sufficient to prevent a shift to a macroalgal forest state. The largest declines in sea urchins at fished sites occurred between 1999 and 2001, and in 2011, and was likely related to blooms of *Ostreopsis siamensis*, a toxic benthic dinoflagellate that blooms in summer and negatively impacts *E. chloroticus* (Shears & Ross, 2009, 2010). The rapid declines at three fished sites between 1999 and 2001 were initially speculated to be due to disease (Shears & Babcock, 2003), but later linked to blooms of *O. siamensis*. Such blooms were not observed at the reserve sites where declines in exposed sea urchins occurred much more gradually (Cole & Keuskamp, 1998, Figure 1). Furthermore, sea urchin populations subsequently recovered at fished sites and, despite some fluctuations in ecosystem state, these blooms did not result in long-term shifts to algal-dominated states. These findings demonstrate how in the fished sites, in the absence of predators, external stressors, such as *O. siamensis* blooms, can result in increased variability in sea urchin populations and greater fluctuations in ecosystem state than in protected areas where predators control urchin populations.

The temporal variation in forest-forming macroalgae and exposed sea urchins was consistent with a discontinuous phase shift pattern between urchin barren and kelp forest stable states (Ling et al., 2015). In the reserve, shifts to kelp forests occurred when exposed sea urchin densities decreased to near-zero (in 2001), yet kelp forests persisted when exposed sea urchins subsequently increased above the initial density in 1999 ($\sim 1 \text{ m}^{-2}$) that were sufficient to maintain barren or turf state and prevent macroalgal forest recovery. The threshold density of sea urchins required to drive a shift from kelp to barrens has not been experimentally determined in this region, but the urchin biomass necessary to achieve this is typically an order of magnitude higher than that required to maintain barrens (Ling et al., 2015). Notably, exposed sea urchin densities have remained comparatively low in the marine reserve despite declines in spiny lobster, one of their main predators (LaScala-Gruenewald et al., 2021). This indicates that predator populations and individual size in the reserve remain at levels sufficient to control exposed sea urchin densities and/or other feedback mechanisms help maintain the kelp forest state (e.g., trophic redundancy [Eisaguirre et al., 2020], behavioral modification in sea urchins [Spyksma et al., 2017]).

There was no evidence of hysteresis between the algal turf state and the other ecosystem states, and the sporadic

occurrence of algal turfs was likely a response to reduced urchin density and/or seasonal variation. The turf-dominated state was only present once in the reserve in the first year of the study and in nine instances at fished sites. In all instances, the algal turf state was short-lived and did not persist in the following monitoring year, except on one occasion when it persisted in two consecutive monitoring years (Nordic: 2001 and 2003). While the infrequent sampling in this study hinders the ability to determine precisely how long this state lasted and whether *O. siamensis* blooms resulted in adverse effects on sea urchins, algal turfs do not appear to form a long-term stable state on these reefs and are instead a successional state associated with reduced sea urchin densities, as supported by sea urchin removal studies (Shears & Babcock, 2002). This contrasts other temperate reef ecosystems where shifts from macroalgal forests to algal turfs may persist as stable states and have been reported as a phase shift (Filbee-Dexter & Wernberg, 2018). In these systems, multiple stressors including long-term environmental changes prevent kelp recovery and promote a persistent turf state independent of sea urchins and their predators. While future environmental changes in our study system that impact kelp and/or sea urchins may increase the persistence and prevalence of turfs, under current conditions sea urchins remain a primary determinant of reef community structure. As a result, persistent and stable kelp forests can establish in the reserve, where sea urchin predators are abundant and can regulate sea urchin numbers and their behavior. In contrast, in the absence of predators due to overfishing, reef ecosystem states can fluctuate along a successional pathway from urchin barrens to kelp forests in response to other factors that result in fluctuations in sea urchin density (e.g., disease, toxic algal blooms, recruitment variability). Ultimately these factors lead to greater variability in rocky reef ecosystems that lack the controlling influence of predators found in the reserve.

These findings have important implications for marine management and conservation by demonstrating that marine reserves can effectively sustain and restore healthy and stable kelp forests in regions where sea urchins are the primary factor influencing kelp abundance. Obtaining a stable kelp forest state, however, may take decades from the time fishing has ceased (Malakhoff & Miller, 2021). The results suggest that should no-take protection be applied today, kelp forest ecosystems may not recover to a desirable stable state until 2050, assuming that predator populations can recover and climatic conditions remain favorable for kelp forests as they exist today. This finding raises questions on the success potential of kelp forest ecosystem restoration as part of the “United Nations Decade

of Ocean Science for Sustainable Development” (www.oceandecade.org) that aims to restore ecosystem health by 2030 (Waltham et al., 2020). Therefore, in order to meet global targets on ecosystem restoration (Duarte et al., 2020; Waltham et al., 2020), it may be necessary to accelerate kelp forest restoration through a combination of marine protection and active restoration at local scales (Eger et al., 2020; Layton et al., 2020; Medrano et al., 2020; Miller et al., 2022). For instance, undertaking sea urchin removal and large predator reintroductions in newly established MPAs could speed up successional recovery and help promote long-term persistence of kelp forests (Johnson et al., 2013; Miller et al., 2022).

Our study demonstrated that the contrasting states between reserve and fished sites have persisted for nearly 20 years despite other stressors, and that overall ecosystem stability of kelp forests in the reserve where predators are abundant was significantly higher than that of urchin barrens in the adjacent fished reefs. The stark contrast in ecosystem state and difference in stability can only be attributed to the effect of protection and the resulting contrast in abundance of large predators between these otherwise environmentally similar locations. This strengthens the evidence for the importance of predators in maintaining stable and resilient kelp forest ecosystems. It also highlights the value of long-established MPAs and long-term monitoring data in understanding the importance of marine protection, and conversely fishing, in structuring kelp forest ecosystems in the face of other anthropogenic stressors.

AUTHOR CONTRIBUTIONS

Nick T. Shears designed and initiated the monitoring program and collected the majority of the data. Caitlin O. Blain and Ohad Peleg contributed to data collection. Ohad Peleg led the conceptualization and writing of the manuscript and performed all analyses. Caitlin O. Blain and Nick T. Shears contributed to the conceptual design of the manuscript and manuscript writing. All authors gave final approval for publication.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data (Shears & Peleg, 2023) are available in PANGAEA: <https://doi.pangaea.de/10.1594/PANGAEA.959778>.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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